

Trainings ▾

Preparatory Steps

QSK45 and QSK60 Single Stage

- Remove the turbochargers. Refer to Procedure 010-034 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>).
- Remove the oil cooler covers. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html>).

QSK60 Two Stage Industrial and Power Generation

- Remove the two-stage platform. Refer to Procedure 010-082 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-082-tr.html>).
- Remove the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>).
- Remove the oil cooler covers. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html>).

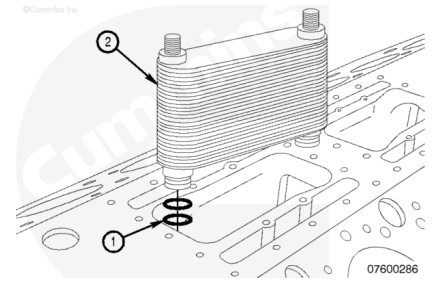
QSK60 Marine Applications

- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>).
- Remove the turbochargers. Refer to Procedure 010-034 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).
- Remove the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-015.html>).
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>).
- Remove the oil cooler covers. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html>).

Remove

⚠ CAUTION ⚠

Do not damage the oil cooler elements by prying with tools that have sharp or pointed edges or corners. The sharp tools can penetrate the oil cooler element and cause a leak.



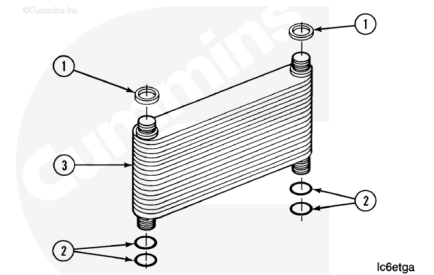
Remove the three oil cooler elements (2) on the QSK45 engine or the four oil cooler elements (2) on a QSK60 engine.

Remove and discard four o-rings (1) from each oil cooler element (2).

Clean and Inspect for Reuse

Note : Cummins Inc. does **not** recommend any form of cleaning of oil cooler elements (internal or external). This includes cleaning with solvent, flushing with oil, steam cleaning and ultrasonic cleaning.

Remove the gaskets (1) and lower o-rings (2) from the cooler element (3).



If the engine experienced a malfunction that allowed debris to pass through the lubrication system to the full-flow filters and oil passages in the lubricating oil cooler become contaminated, the oil cooler elements **must** be replaced.

Inspect the oil cooler elements externally for scale build-up, corrosion, embedded debris, and/or any damage to the element. If there is any evidence of any of these conditions, the oil cooler elements **must** be replaced.

Note : Contamination can be defined by metal, rubber, or other solid particles present in the lubricating oil filter and/or by debris damage to the bearings, or debris damage of another component, such that they **do not** meet reuse guidelines.

Remove gasket material from the oil cooler cover, oil cooler element, and engine block mating surface.

Use solvent to clean the oil cooler cover.

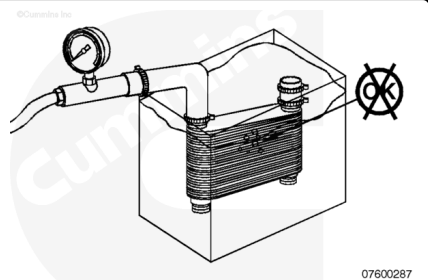
Dry the oil cooler cover with compressed air.

Clean the covers with solvent.
Inspect the covers for corrosion.



Pressure Test

Plug the oil cooler element as illustrated and connect to a source of compressed air.
Pressure-test each oil cooler element to check for leaks.



Measurements

	kpa	psi
Air Pressure:	517	75

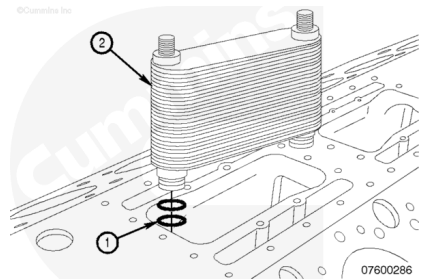
Note : Heating the water in the tank to 50°C [122°F] will improve the test results.

If the oil cooler element leaks, it **must** be replaced. Do **not** attempt to repair an oil cooler element.

Install

⚠ CAUTION ⚠

All water or cleaning solutions must be removed from the interior of the oil cooler element. A failure can result from a contaminated lubrication system if all of the liquid is not removed.



⚠ CAUTION ⚠

When installing new oil cooler elements, make sure that all of the shipping plugs have been removed from the oil cooler elements or the engine will be severely damaged by lack of lubrication.

Install the four o-rings (1) in the grooves on each oil cooler element.

Lubricate the four o-rings (1) with vegetable oil for each oil cooler element.

The QSK45 engine contains three oil cooler elements and the QSK60 contains four oil cooler elements.

Install the oil cooler elements.

Install the support gaskets on the top post of the oil cooler element.

Finishing Steps

QSK45 and QSK60 Single Stage

- Install the oil cooler covers. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html>).
- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>).
- Install the turbochargers. Refer to Procedure 010-034 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).

QSK60 Two Stage Industrial and Power Generation

- Install the oil cooler covers. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html>).
- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>).
- Install the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).
- Install the two-stage platform. Refer to Procedure 010-082 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-082-tr.html>).

QSK60 Marine Applications

- Install the oil cooler covers. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html>).
- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>).
- Install the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11 (</qs3/pubsys2/xml/en/procedures/56/56-011-015.html>).
- Install the turbochargers. Refer to Procedure 010-034 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>).

Last Modified: 26-Apr-2016
